How to preprocess images

RegiStar allows you to perform common preprocessing tasks to correct images for hot pixels, vignetting, and dust. Various categories of images, as itemized below, may be used depending on the characteristics of the camera and optics. Images in categories #2, #3, #4, and #5 are applied as corrections to the base images in category #1. All images are taken with the same camera and image-acquisition software, and the same optical configuration must be used for #1 and #3. All images must contain linear pixel data, which for example is the case for raw images.

1. *Normal images* containing actual image data that is to be corrected. Ideally, all images are taken with the same optical configuration, same sensor temperature, and same exposure time.

2. *Dark frames* containing hot-pixel data. The images have the same exposure time and sensor temperature as those in #1. The same set of dark frames can be used for all corresponding sets of normal exposures. The camera is capped so that no light reaches the sensor.

3. *Flat fields* that record light fall-off caused by vignetting and dust. The same optical configuration as in #1 is used. The same set of flat fields may be usable for several sets of normal exposures. However, if you remove the camera between exposures, you may want to acquire another set of flats to account for possible changes in camera orientation or sensor dust. Short exposures are taken of an even, bright target.

4. *Dark-flat frames* containing hot-pixel data for the images in #3. These images may not be necessary if the exposures in #3 are very short. As in #2, the camera is capped.

5. *Bias frames* containing base camera readout. These exposures are very short. Bias frames are sometimes subtracted from flat-field frames. Also, when subtracted from images in #2, the resulting bias-corrected dark frames can be extrapolated to other exposure times, so that dark frames need not have the same exposure times as the normal images.

RegiStar does not currently fully support #5 (bias frames). However, note that since dark frames can be subtracted from flat-field frames, you can subtract a bias frame from a flat-field frame by designating the bias as a dark. Also, see the paragraph below regarding the second use of bias frames mentioned in #5. Otherwise, images in #2 should, as stated, match the exposure times of those in #1.

The other three corrections listed above are fully supported. Using the Open command, images in #2, #3, and #4 can be tagged respectively as dark frames, flat fields, and dark frames, which allows them to then be applied as preprocessing corrections to normal images in #1. Images in #2 and #4 should not be opened at the same time to avoid mixing them.

Typically, several exposures are taken in each category, to mitigate random noise in individual images. The Combine command can be used to produce a "master" image in #2, #3, and #4 by doing some sort of average of similar individual images. In the Combine control, you can use the *Function to apply* section to specify a particular averaging option. You can also add or subtract images with weights, which allows you to produce extrapolated dark frames mentioned in #5. Averaging can be done on a collection of dark frames or flat fields (see below), but extrapolation cannot.

The Preprocess command can then be used to apply corrections #2 and #3 to base images in #1 and also corrections #4 to images in #3. A typical workflow involving just #1, #2, and #3 would be:

a. Place a set of normal (#1) images into directory `A`.

b. Place a corresponding set of dark frames (#2) into directory `B`.

c. Place a corresponding set of flat fields (#3) into directory `C`.

d. Open all images in directory `B`, using a single Open command with all images in the directory selected and with the dark-frame image type selected.

e. Do a Combine/average on the images opened in step d to produce a master dark, which will be called `Combine1`.

f. Open all images in directory `C`, using a single Open command with all images in the directory selected and with the flat-field image type selected.

g. Do a Combine/average on the images opened in step f to produce a master flat, which will be called `Combine2`.

h. Select `Combine1` and use a single bulk Preprocess command to subtract this master dark from all images in directory `A`, specifying that the resulting images are to be saved in directory `H`.

i. Select `Combine2` and use a single bulk Preprocess command to divide this master flat into all images in directory `H`, specifying that the resulting images are to be saved in directory `I`.

Directory `I` will then contain dark-subtracted and flat-fielded versions of the original normal images, which can be processed and assembled using Register, Combine, and other operations. Six RegiStar commands are required above: two Opens, two Combines, and two Preprocesses. If you only want to perform dark subtraction, omit steps c, f, g, and i; the final dark-subtracted images will be in directory `H`. Note that for flat-fielding (step i) to work properly, you must first perform dark subtraction (step h).

If any of the images are raw, they must all be and must all be opened as such. For camera-raw images, that will happen automatically. FITS-raw images must be declared to be raw. Using a Raw-parameters file is a convenient way to do so; see also Raw parameters. Note, as explained in Preprocess, that it is not necessary for all images to have the same color multipliers, but they must have the same Bayer array.

Scripting is a convenient alternative to the interactive workflow above, as it can completely automate this sort of repetitive work. See Sample scripts for an example of a short preprocessing script that performs the workflow above and which can be easily edited for different image sets.

How to register images

Once any necessary preprocessing has been done, the first step in assembling various images is to register them so that they are properly aligned. This step must be done before other operations such as Combine and Merge to RGB can be performed to produce final assembled images.

To register one image to a second one, open both images and select the one that you want to register. Run the Register command on that image, and specify in the Registration control dialog the other image's group as the *Reference Group.* Initially, each image is put into a group consisting of a single image, with the name of the group being the name of the image without the suffix.

As the Register operation proceeds, a new image is created from the first image that is registered to the second image, as well as to all other open images in that second image's group. This new image is placed in the same group as the second image so that these images can later be assembled using another RegiStar operation.

To register several images to each other, select one as the base image and register one of the other images to it. All subsequent images have the same resolution and orientation as this base image. Then, successively register the other images to the group of this base image. This group grows in size as more images are registered to it. Every image in this group is registered to every other open image in the group with which it has some overlap. The number of Register operations required is one less than the total number of images.

The *Multiple source* option in the Register command is a convenient way to register several images to each other. Open one of the images and select it as the base image; if the other images all reside in a common directory then you can select the *Multiple source* option in the Registration control to instruct RegiStar to progressively register these other images to the selected reference image, creating a group of registered images as above.

If you are registering several images to each other, and these images have average star densities that vary

considerably between the images (by a factor of 2 or more), you will generally get the best results if you use the image with the greatest star density as the base image and then successively register the other images in order of decreasing star density. RegiStar usually registers images successfully regardless of the order in which they are registered; slightly better accuracy may be achieved if the images are ordered as suggested.

To reduce memory consumption, you can close the source for an image after it has been registered. See Hints on reducing memory usage for other tips on reducing memory requirements.

RegiStar's scripting ability lets you use a script to automate long sequences of operations. See How to use automated scripts for instructions and sample scripts.

How to stack images

Digital stacking is the process of combining two or more exposures of the same object. Various exposures of the same field, when combined, generally yield a superior image to the individual components since the resulting final image has less noise.

To stack a set of images:

1. Preprocess the images as necessary.
2. Adjust raw parameters of any raw exposures as necessary. See the *Assigning Raw Parameters to Image Sets* section of Raw-image hints.
3. Select one of the images (preprocessed, if step 1 was performed), which will serve as the *base image* for the stack.
4. Adjust the base image's orientation, if you wish, using the Rotate/Flip command. This step is entirely optional, since the Register command automatically handles rotation and geometric inversion. You can instead correct the orientation of the final processed image.
5. Register these images. If all the images are in a single directory, you can use the *Multiple source* option.
6. Use the Combine command to stack the registered set of exposures.

The Calibrate command can be used to compensate for differences in brightness or contrast between images after they have been registered and before stacking them.

If you want to stack the registered images using another program instead of using RegiStar's Combine command (step 6), you will need to equalize their dimensions. You can do so in step 5 by using the *Registered image dimension specification* section in the Registration control, or by using the Crop/Pad command afterwards.

While it is traditional to use one fixed optical system for stacking images, you can use RegiStar's image deformation capabilities to stack images taken with two or more different optical systems.

RegiStar's scripting ability lets you use a script to automate long sequences of operations. See How to use automated scripts for instructions and sample scripts.

How to form mosaics

Mosaics are images formed by combining exposures of two or more overlapping but distinct frames. They can have considerably higher resolution than a single-frame image of the same total field taken with optics that have a shorter focal length.

To form a mosaic from a set of exposures:

1. Preprocess the images. Flat-field correction is usually required for mosaicking but can sometimes be omitted if the camera sensor is not large.

2. Adjust raw parameters of any raw exposures as necessary. See the *Assigning Raw Parameters to Image Sets* section of Raw-image hints.

3. Select one of the images (preprocessed, if step 1 was performed), which will serve as the *base image* for the mosaic. For best results, choose an image that is in a central frame.

4. Adjust the base image's orientation, if you wish, using the Rotate/Flip command. This step is entirely optional, since the Register command automatically handles rotation and geometric inversion. You can instead correct the orientation of the final processed image.

5. Register all images to the selected base image, using the *Multiple source* option. Name or organize the images so that in their lexicographical order later images always have an overlap with earlier ones. One way to organize is is to use a single folder and name the images using the approximate coordinates of the image's centers. Another way is to use frame-by-frame subfolders of a folder that you specify with the *Include subdirectories* option in the Registration control. Name the subfolders in a spiral starting with the folder containing the base image. For example, for a 3 x 5 mosaic, you could name the 15 frame subfolders:

   ```
   15 11 02 06 12
   10 05 01 03 07
   14 09 04 08 13
   ```

6. Use the Calibrate command to remove differences in brightness or contrast between images. Select the *All open images in the current image group* option, which will calibrate *all* group images that have an iterated overlap with the selected reference image, not just those with a direct overlap. Before applying the command, close any images that will not actually be used to form the final composite image, and use the *Close original image(s) after calibration* option to only leave calibrated images open.

7. Using the Combine command, amalgamate the registered exposures. The *Max* function to apply option can hide seams if images have not been flat fielded, but can also lead to some degradation of image quality. If unneeded images have been closed as suggested in step 6, you can use the *Select all images* option without subsequently deselecting individual images.

You can use RegiStar's image-deformation capabilities to form mosaics from exposures taken with two or more different optical systems. For example, the southern mosaic above used both a Takahashi FSQ refractor and a Celestron Rowe-Ackermann Schmidt Astrograph.

If a large number of frames is involved, peripheral regions far from the base frame may exhibit a large amount of geometric distortion. Consider instead using as the base image in step 2 above a wide-field image taken with a shorter focal length, or even output from a star map that spans the desired mosaic. Use a general-purpose imaging program to sample this wide image up to the desired resolution and then register the original exposures to it. If you use this method, you do not need to order the individual frames as described in step 5 above. *The wide-field base image is used solely to define the geometry of the mosaic; it should* ***not*** *be included in steps 6 or 7.*

RegiStar's scripting ability lets you use a script to automate long sequences of operations. See How to use automated scripts for instructions and sample scripts.

How to form tricolor images

Tricolor images are color (RGB) images formed by combining 3 separate grayscale images taken through red, green, or blue filters respectively. Since monochrome cameras generally have better resolution than color cameras, tricolor images are usually superior to standard color images. Also, tricolor images may have more accurate color information.

To form a tricolor image from a set of three (R,G,B) exposures:

1. Preprocess the images as necessary.
2. Register the images (or select one of the images and use the *Multiple source* option).
3. Using the Merge to RGB command, assemble the registered tricolor set into a color image.

While it is traditional to use one fixed optical system for tricolor image formation, you can use RegiStar's image deformation capabilities to form tricolor images by using exposures taken with two or more different optical systems. Also, each of the individual R,G,B images can itself be a stack of two or more different grayscale exposures.

RegiStar's scripting ability lets you use a script to automate long sequences of operations. See How to use automated scripts for instructions and sample scripts.

Execute/Resume Script

Available in the File menu, the **Execute Script** command instructs RegiStar to run a list of commands, comprising a *script*, one after the other. Select the script in the **Open Script File** dialog.

By invoking this command, you can have RegiStar automatically run a specified sequence of operations, without any intervention, and inspect the results after the script has completed. You can also instruct the script to pause at specified points so that you can study intermediate results.

A script enters suspended mode when a Suspend command in the script is executed. If a script is suspended, you can resume execution of the script using the **Resume Script** command. The File menu displays the Resume Script command if a script is suspended and the Execute Script command if no script is running.

When a script is executing, its file name and status — *Running* or *Suspended* — are reported in the upper status bar. You cannot start another operation while a script is executing, even if it is suspended. However, if a script is suspended then some dialogs become available.

The sequence of commands must be saved in an ordinary text file which you can specify in the *Open Script File dialog.* See Script files, Scripting commands, and Scripting directives for instructions on writing this script file.

When a script is run, it passes through two phases. First, RegiStar validates the script without actually executing it, checking for errors and that the directories and files referred to in the script exist. If the script passes this validation, it proceeds to the execution phase. You can specify in the script that it is to be executed if there are no validation-time anomalies other than missing files. If you choose this option, you must ensure that the necessary files exist by the time they are encountered by the script. Execution of a script stops when a missing file is encountered.

Confirmation dialogs that appear when a command is run interactively — namely when a large image is about to be created or when a large number of images are about to be opened in bulk — are skipped when that command is run in a script. An out-of-memory crash or long uninterruptible operation may result if a script includes a runaway command.

A log file (.log) in the same directory as the script file and with the same base name is created whenever a script is executed. It records any anomalies encountered during the validation or execution phases. If a script fails validation or does not yield the expected results after it is executed, you should inspect this file. You can view the log file by using any text editor after the script's execution is complete. Some text editors, such as *Notepad* or *Word,* allow you to view this file while the script is running. Other text editors such as *WordPad* may refuse to open the log file while the script is running and that file is being used by RegiStar.

You should ensure that scripting commands that require images to be in the same group satisfy this grouping requirement, as consistency of grouping is not checked during the validation phase. If during the execution phase

a command is encountered that fails required grouping, execution of the script stops. If a command cannot be completed for some other reason, for example a requested registration cannot be performed or an image about to be created is too large to fit into available physical or virtual memory, a warning to that effect is recorded in the log file but script execution proceeds nevertheless. However, the script may subsequently terminate with an error if the image that was to have been produced by this incomplete operation is used later in the script.

If an error is encountered while a script is executing, a cautionary message box appears. This message disappears after 20 seconds or when the *OK* button is pressed.

You can stop a script by using the Stop command. This stops the current operation being executed. If you stop a script, you cannot re-enter it midstream. Instead, you must rerun the script in its entirety. You can pause an operation being run by a script, by using the Pause command.

You can fully inspect all open images while a script is in progress but you cannot close any image opened by or produced by the script. You must stop the script first. If you attempt to close an image involved in an active script, a message appears and the request is ignored.

Scripts can also be run using a command line or from within a batch file. See Command-line scripting for details.

Shortcut

Key: CTRL + E